

# Summarizer Training Report

Qwen3-4B-Instruct fine-tuned on CNN/DailyMail with QLoRA | RTX 3060 (12 GB) | October 4, 2026

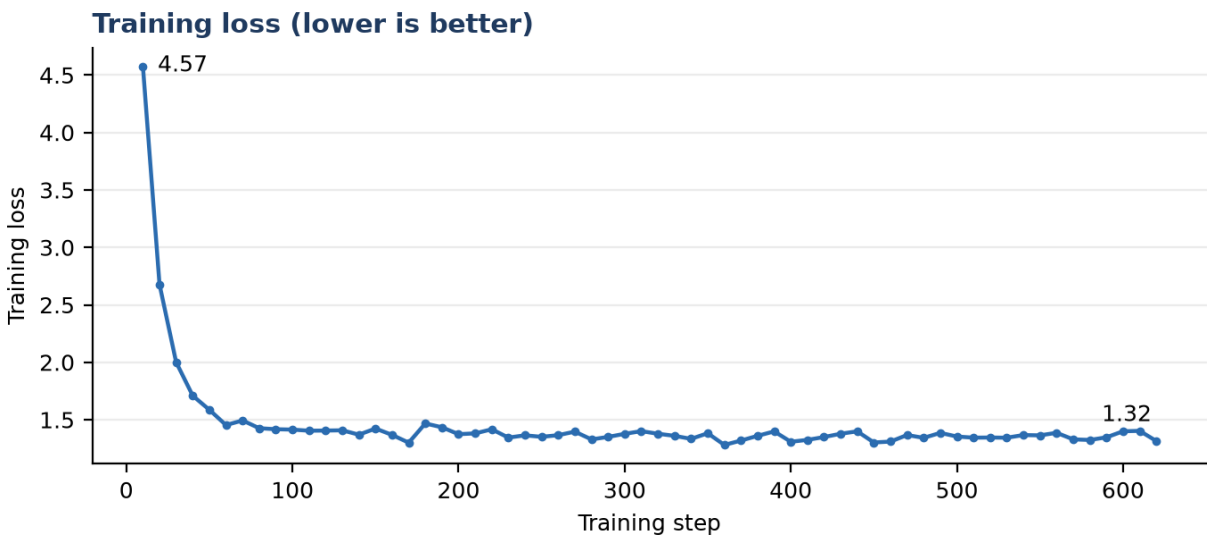
## Result at a glance

The fine-tuned model beats the untuned base model on every ROUGE metric on the same 200 test articles. ROUGE-1 rose from **34.03** to **39.89** (+5.86) and ROUGE-Lsum from **30.33** to **37.43** (+7.10). Validation loss fell from 1.363 (step 100) to **1.293** at the end of training.

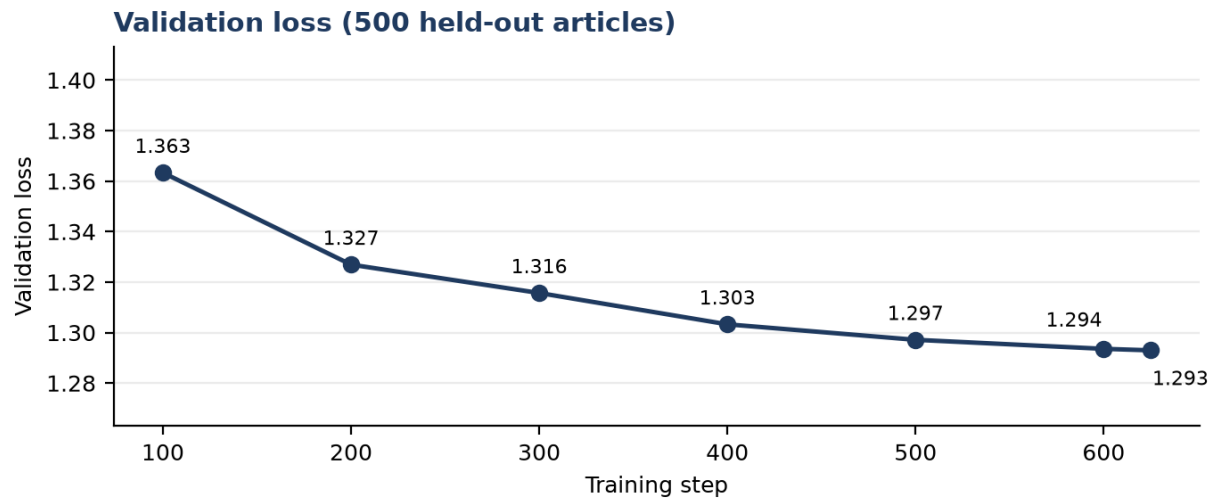
## Training setup

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Base model                     | unsloth/Qwen3-4B-Instruct-2507 (4-bit, QLoRA)                                               |
| Dataset                        | abisee/cnn_dailymail 3.0.0 (news article to highlights)                                     |
| Training / validation examples | 10,000 / 500 (articles over 2,048 tokens removed)                                           |
| Trainable parameters           | 33,030,144 of 4,055,498,240 (0.81%), LoRA rank 16                                           |
| Batch size                     | 2 per step x 8 accumulation = 16 examples per update                                        |
| Learning rate / optimizer      | 2e-4, linear decay, 30 warm-up steps / AdamW 8-bit                                          |
| Epochs / total steps           | 1 epoch / 625 steps                                                                         |
| Precision                      | bfloat16                                                                                    |
| Hardware                       | NVIDIA GeForce RTX 3060, 12 GB, Windows (native)                                            |
| Speed                          | about 20-22 seconds per step; one validation pass takes about 3.6 minutes                   |
| Run history                    | Ran to step 600, was interrupted, then resumed from checkpoint-600 and finished at step 625 |

## Training curves

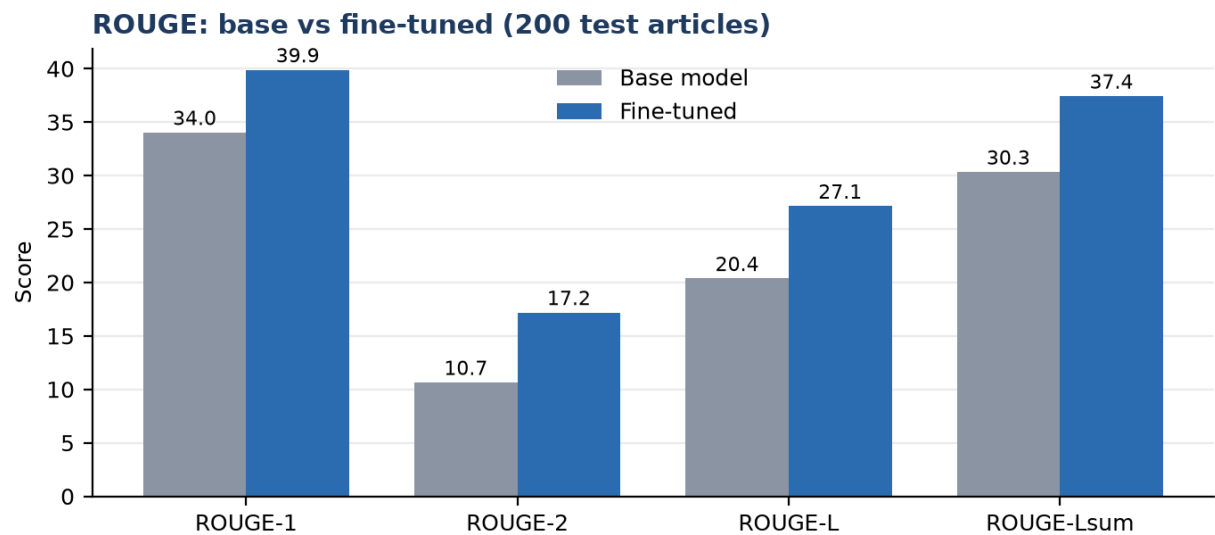


Loss dropped fast in the first 30 steps (4.57 to about 2.0) and then settled near 1.3-1.4.



Validation loss improved at every check, from 1.363 to 1.293, with smaller gains later. It stayed close to the training loss, so the model is not memorizing the training data.

## ROUGE scores: base vs fine-tuned



| Metric     | Base  | Fine-tuned | Change |
|------------|-------|------------|--------|
| ROUGE-1    | 34.03 | 39.89      | +5.86  |
| ROUGE-2    | 10.65 | 17.19      | +6.54  |
| ROUGE-L    | 20.37 | 27.14      | +6.77  |
| ROUGE-Lsum | 30.33 | 37.43      | +7.10  |

Both models were scored on the same 200 randomly chosen test articles (fixed seed), using greedy decoding with up to 200 new tokens. An earlier 50-article check of the fine-tuned model gave ROUGE-1 38.21 / ROUGE-2 15.19 / ROUGE-L 25.86 / ROUGE-Lsum 36.06. With only 200 articles, scores can shift by about 1-2 points between samples, but the gap between the two models (5.9-7.1 points) is much larger than that.

## Summary length

|                              | Avg. words per summary |
|------------------------------|------------------------|
| Reference (human highlights) | 55.3                   |
| Fine-tuned model             | 64.2                   |
| Base model                   | 96.7                   |

## Example: summary of an unseen article

**Article:** "United States announces blockade on the Strait of Hormuz" (Wikinews, CC BY), about 250 words.

### Fine-tuned model output

President Donald Trump announces blockade on the Strait of Hormuz .  
Blockade follows collapse of talks between the U.S. and Iran .  
Oil prices rise above US\$100 a barrel, then ease back to just over US\$99 .

The three sentences match the article's main facts: the blockade, the collapsed talks, and the oil price move. No invented details were found. The space before each full stop is the dataset's own style.

## Files produced

|                                             |                                                    |
|---------------------------------------------|----------------------------------------------------|
| outputs/qwen3-4b-cnndm-lora                 | Final trained LoRA adapter (use with summarize.py) |
| outputs/eval_base.json, eval_finetuned.json | ROUGE scores plus every generated summary          |
| outputs/checkpoints/checkpoint-625          | Last training checkpoint with the full loss log    |

## Notes

- ROUGE measures word overlap with the reference summaries, not truthfulness. Read a sample of summaries to judge quality.
- The model was trained on English news, so other text types and languages will give weaker results.
- Given an empty input, the model invents a plausible-sounding story. Always check that the input file is not empty.
- The training log's final "train\_loss" value (0.054) is an artifact of resuming and should be ignored. The per-step losses above are the correct ones.